

Antimicrobial Resistance (AMR) Beyond the Diagnosis

*“A **sinking** feeling is what you feel first: I’ve had this before and oh no, it’s happening again.” - Norma Washburn, AMR patient*

*“[My] dressing lasted for months, [but I was] **scared** it (AMR) would happen again.”*

*“[But it’s a] **relief** to have people who are specialists to do these dressings for you”*

*Initial **hope/happiness**, then **despair** but then **encouraged** and ultimately back down in the “**dumps**.”*

These are the rollercoaster of emotions familiar to patients with antimicrobial resistance (AMR). These are real feelings experienced by real patients, anecdotes that are not impossibly distant from anyone reading this narrative. This subconscious despair can even manifest itself physically, where a patient collapses to the ground with exhaustion after repeated treatment with antibiotics.

Getting the right antibiotic matters. But for someone living with an antimicrobial-resistant infection, treatment is only part of the story. In investigation of what this means for patients, we have spoken with physicians, nurses and other healthcare providers as well as patients, having learned that some of the hardest parts of antimicrobial resistance (AMR) can happen outside of the hospital, in a place much more personal and tangible: One’s spirit.

AMR can change how patients receive care, how others interact with them and even how they see themselves. It is important to understand the aforementioned in order to develop effective healthcare implementations that support the whole of patients.

The Problem

Patient-physician Interactions

“I want others to understand how I feel and stop long enough to allow me to share how I feel instead of going straight into...the treatment” - Sunny Loo, vasculitis patient

“I would appreciate [my physician] asking about how I feel...[but] they would send me straight to counseling if I did ask,” adding that the cost for this, even, would be out of pocket.

Physicians are often the first people who break the news of conditions such as AMR to their patients. Having sworn to do no harm and serve humanity, physicians choose their profession out of a deep passion for helping mankind—one that transcends far beyond words. Unfortunately, in light of the major physician shortages across Canada (including British Columbia), it might be difficult for healthcare providers to find the time to connect with patients more personally. Regardless, this is an important aspect of inclusive care, as voiced by our patient Norma:

“It seems they have just decided that their priority is getting through patients after patients, but it leaves the patient feeling less than prepared to face what they need.”

“Make it a priority to, if the doctor has concerns about a medication and is hovering between several options, [tell] the patient...that there are those concerns...he should let the patient in on it so that they know what will be possible outcomes.”

She added that her physician often feels rushed during their consultations and does not have the time to explain to her what AMR is, what side effects she can expect and maintaining overall transparency about treatment options. This made the later comorbidities associated with her treatment an unexpected phenomenon that could have been addressed earlier, had this conversation taken place. This point is further underscored by **Dr. Anthony Liu**, mentioned that as physicians move higher up in their training, they might lose earlier knowledge about the basics of certain diseases and

thus, not explain this adequately to their patients. Additionally, **Lesley Kooner**, an infectious control nurse and former bedside nurse, extends this to nurses, supporting Dr. Liu in a later assertion made by him that nurses—being the healthcare providers that spend the most time with patients—should also be well-versed in AMR so they can reciprocally educate their patients.

In fact, the lack of the matter feeds into a psychosocial restraint experienced by AMR patients when they have limited knowledge about AMR screening protocols:

“They (nurses) are busy putting in a PICC telling me that I need it, but only because I have some nursing experience I [understood] it...I would be really really scared [otherwise]...people have no time to explain to patients”

Dr. Edith Blondel-Hill mentioned in an interview with us that some of the screening protocols for AMR can be invasive to patients’ sense of privacy and modesty, such as anal swabs or repeated swabs of other potentially infected areas. Without sufficient debriefing before the screening procedure, the patient can feel violated and experience psychological strain.

“I wish I had an in-person consultation but I also wish that I had looked up some of these things myself” - Norma, AMR patient

Norma also brings to light the importance of patients taking a leading role in their own treatment process, something endorsed by Lesley Kooner and **Dr. Richard Lester** (infectious disease physician).

Isolation

For some patients, one of the first visible changes may be hospital staff, family members, and friends suddenly needing gowns, gloves or other protective equipment before visiting their room. While these measures are important for preventing resistant bacteria from spreading, they can change the patient's experience of care.

Our conversations highlighted how easily physical isolation can become social isolation. A patient may begin wondering whether they are dangerous to the people around them.

Some distance themselves from partners or family members because they fear passing the infection on. Others can begin to perceive themselves as “dirty” or contagious. Infectious disease physicians such as Dr. Edith Blondel-Hill, Dr. Anthony Liu suggested that isolated patients may receive lower-quality care. This is supported by empirical evidence, showing that these patients receive fewer or shorter healthcare interactions and can experience anxiety, depression and stigmatization ¹.

Norma described to us that the church and her friend who believes in the power of prayer has helped her greatly in times of isolation, and Sunny described to us that he enjoys playing online games with friends to continue connecting with others. This shows that even in isolation, patients can help alleviate their conditions by companionship or spiritual support.

Unfortunately, there is also a lack of community amongst AMR patients, raised by both Sunny and Norma, mentioning that they would appreciate the presence of such a community to connect with.

Educational Shortcomings

One theme came up repeatedly in our conversations: Patients often don't fully understand what AMR means. A common misconception is that the person has become resistant to antibiotics. They haven't; the bacteria have.

“Mom, remember when we were kids and you were reading us all about the fact of overuse of antibiotics and [that] we’re gonna create a superbug? Mom, this is what you got - a superbug!” - Norma’s daughter

Others hear terms such as “superbug” and assume that no antibiotic will work at all; these terms came up in our patient interview as well as educational initiatives with high school kids. It is also shown in the literature that this is a term often misused by

patients ². In reality, clinicians often have alternative treatment options, but resistance can make those options more limited and treatment more complicated. This confusion is why education isn't an optional extra in AMR care. Patients need opportunities to

understand what they have, why a particular antibiotic was selected, what precautions mean and what they can do to participate in their own care.

Doctors [get straight to] what illness, what procedure, etc... but often this is a small part of a patient's day to day concern...most patients require a psychological aspect, how do I go through the rest of the day with what I have, how do I get through tomorrow with what I have?" - Sunny Loo, vasculitis patient

But education cannot stop with patients. As doctors become more specialized in their practice, they often lose the basics and end up debriefing patients less, as previously noted by Dr. Liu. And in a very busy healthcare setting where there can be a culture of patients not wanting to bring issues up, patient understanding can be overlooked. Nurses also spend significant amounts of time interacting with patients and can play an important role in both education and antimicrobial stewardship. Our conversations highlighted the importance of continuing to educate healthcare providers about AMR, appropriate testing and recognizing the right signs and symptoms before automatically moving toward testing or treatment.

Treatment has to Fit Life

Eventually, the patient leaves the hospital room and the infection goes with them, but so does everything else in their life: work, school, dependents, relationships, and finances. A resistant infection can mean repeat appointments, additional testing, specialist visits and longer treatment. Someone living close to a major hospital may be able to attend those appointments relatively easily. Someone living in a smaller community may not have the same access to specialized care.

And every appointment can carry another cost. This could mean the mental strain involved in rearranging another week around treatment. It could also mean the exhaustion of having to make yet another journey back and forth. Norma reported that she once collapsed from going to the hospital so much.

Even the form of treatment matters. Some IV treatments require resources such as refrigeration, which may not be practical for every patient. Clinicians may therefore

consider oral medications or longer-lasting treatments when appropriate to reduce the number of hospital visits a patient needs to make. Unfortunately, the best treatment on paper may not be the best fit for somebody's life.

How can we improve AMR care?

"We fail to address the whole person." - Norma, AMR patient

Preventing and treating AMR infections requires better diagnostics, effective antibiotics and responsible antibiotic use. But our conversations suggest that good AMR care also requires communication, education and empathy. Patients can experience anxiety, stigma and fear of transmitting resistant infections to the people around them. For people undergoing prolonged treatment, those psychosocial effects can exist alongside repeated medical procedures and medication side effects.

The amalgamation of our findings translate into the following categories with associated action items:

- **To physicians; consider these:**

- Explaining isolation precautions before they happen and that it is meant to keep the patient and others safe
- Reassessing whether isolation is still necessary when appropriate
- Checking in after an uncomfortable screening procedure
- Making sure a patient understands the course of their treatment, including potential side effects and how AMR works
- Maintaining transparency about different treatment options and sharing with the patient your thought process for those treatments - giving patients a role in their own care
- Maintaining updated knowledge about the side effects and mechanisms behind AMR and antibiotics

- **To nurses; consider these:**

- Debriefing the patient before screening procedures, especially those that can be invasive to a patient's sense of modesty

- Ensuring patients understand the treatments you administer to them
- Keeping your knowledge up to date with the literature regarding the mechanisms behind AMR
- **To patients:**
 - Take a lead in your treatment and ensure you understand
 - Seek spiritual or companionship support in the form of virtual interactions, should the patient be isolated

“When we talk about our experience, we are just asking to be understood, to be supported.” - Sunny Loo

AMR isn't only about whether a drug can kill a bacterium; it is about a “whole person’s” journey in fighting off a straining condition that can put their social, psychological and spiritual health on a tangent.

Works Cited

Crago, A.-L., Alexandre, S., Abdesselam, K., Gravel Tropper, D., Hartmann, M., Smith, G., & Lary, T. (2022). Understanding Canadians’ knowledge, attitudes and practices related to antimicrobial resistance and antibiotic use: Results from

public opinion research. *Canada Communicable Disease Report*, 48(11/12).
<https://doi.org/10.14745/ccdr.v48i1112a08>

Mellinghoff, S. C., Grossi, A. A., Recanatini, C., Breull-Wierschem, L., Salm, F.,
Gadebusch-Bondio, M., & Jung, N. (2026). The human cost of resistance: ethical
implications of coping with isolation for multidrug resistant organisms. *Clinical
microbiology and infection : the official publication of the European Society of
Clinical Microbiology and Infectious Diseases*, 32(8), 1244–1249.
<https://doi.org/10.1016/j.cmi.2026.05.043>